Skin Lesion Diagnosis: Classification of Dermal Medical Imaging

Goal Statement

Be able to distinguish between different pigmented skin lesions using dermatoscopic images and metadata with an accuracy of above 90%.

Research Question:

Can we develop a model that distinguishes between common pigmented skin lesions using dermatoscopic images?

Modeling Approach:

For our modeling approach, we want to be able to correctly identify the type of skin lesion. Our dataset includes both images of skin lesions and additional patient information, so we will use a multimodal approach that combines both types of data. Before we build the model, we will preprocess the data. This includes making sure all the images are the same size, and cleaning the metadata by handling missing values and/or deleting unnecessary rows. For the model, we want to use a CNN to extract key features from the images [2]. Then we will train the images and metadata, which will then be combined and run through a classifier. We will evaluate the model using precision, recall, f1, and accuracy. We will then repeat the same process for a model without additional patient data.

Executive Summary

This document outlines our dataset establishment details and analysis plan for skin lesion predictive diagnoses using image data modeling.

Dataset Establishment Details

Our data is gathered from a dataset on the Harvard Dataverse called "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." The dataset in the Harvard Dataverse contained several different files, with the data itself being split up into two files with roughly 5000 JPEG files each. We used the two files containing the images as well as the file containing metadata about the images. The dataset was very clean to begin with so the only cleaning we did was combining the two parts, dropping unnecessary columns (not included in the data dictionary), and filling null values in each column. We also added a new column which added the image path called image_path to the metadata, so all of the information we would need could be located in one place.

Data Dictionary:

[33]:

	lesion_id	image_id	dx	dx_type	age	sex	localization	dataset	image_path
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	vidir_modern	data/images/ISIC_0027419.jpg
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	vidir_modern	data/images/ISIC_0025030.jpg
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	vidir_modern	data/images/ISIC_0026769.jpg
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	vidir_modern	data/images/ISIC_0025661.jpg
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	vidir_modern	data/images/ISIC_0031633.jpg

	image_path	dx	age	localization_abdomen	localization_acral	localization_back	localization_chest	localization_ear	localization_face	localiz
0	data/images/ISIC_0027419.jpg	2	80.0	False	False	False	False	False	False	
1	data/images/ISIC_0025030.jpg	2	80.0	False	False	False	False	False	False	
2	data/images/ISIC_0026769.jpg	2	80.0	False	False	False	False	False	False	
3	data/images/ISIC_0025661.jpg	2	80.0	False	False	False	False	False	False	
4	data/images/ISIC_0031633.jpg	2	75.0	False	False	False	False	True	False	

Column	**Description**
lesion_id	How the images are tracked
image_id	The ISIC image id of the image during study
dx	The diagnostic category of the lesions
dx_type	The mode in which the lesion was tested
age	The age of the patient
sex	The sex of the patient
localization	The location of the lesion on the patient
dataset	The origin source of the image
image_path	The location of the image

Our initial question when beginning this project was: Can we build a system that accurately identifies different skin spots using close-up photos and census info at least 90% of the time?

We explored the distribution of the dataset by lesion location and type of lesion and found that the events were skewed towards back and lower extremities as well as nv (melanocytic nevi—a benign skin lesion). We also explored the distribution of age and found that the data is skewed slightly to the right, with the most concentrated ages being early 40s. The split of male to female patients was fairly even, with roughly 10% more males. We were interested in whether the sex and age of the patient affected the class of lesions. One of the classes of skin lesions for example, bcc (basal cell carcinoma), presents more frequently in males over the age of 60 so we created

two heat maps to look into potential relationships. The first heatmap compared age with lesion class and we found that most of lesions were found between the ages 35-70. The second heatmap compared sex with lesion class and we found that the spread of lesions was fairly even between males and females. The heatmaps indicate that there could be a relationship between age and lesion class but likely not a relationship between sex and lesion class, so we decided to include age but not sex in our model.

We were interested in whether the localization of the lesion affected the type of lesion. We created a heatmap that compared lesion localization with lesion class and found that the majority of the nv lesions were located on parts of the body frequently exposed to the sun, such as arms and legs. The nv lesions are benign moles, primarily caused by sun exposure [4], so the heatmap indicates that there might be a relationship between lesion localization and lesion class. Thus we will include the lesion localization metadata in our model.

After doing EDA and looking into potential relationships between lesion class and different aspects of the metadata, our next question is: Will a model perform better with or without metadata? We will explore this question as well as our initial question by running two models: one with the inclusion of the metadata and one without.

Analysis Plan

Step 1: Find Data and Preprocessing

- Search through approved data source locations to find a dataset with respect to images data. Once final selection is made, begin preprocessing. We cleaned the data as discussed in the Dataset Establishment Details section.

Step 2: Data Analysis

- We conducted EDA to understand the structure and characteristics of the dataset. We examined distribution of lesion types, patient age, sex, and lesion localization, and generated heatmaps comparing each metadata feature against lesion class. We found the dataset is heavily imbalanced, confirming the need to address class imbalance during training. Based on EDA findings, age and localization were selected as metadata features for the model, while sex was excluded.

Step 3: Training Strategy, Data Splitting, and Image Extraction

- Before model training, the data will be split using Stratified K-Fold Cross-Validation with 5 folds. The model will then be trained and evaluated across all 5 folds. After splitting and training the data, image features will be extracted using ResNet50 and MobileNetV2, two CNN models pre-trained on ImageNet.

Step 4: Model Development

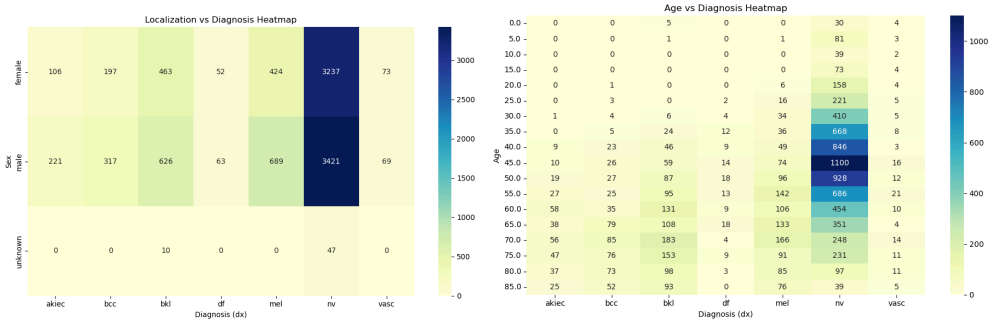
- We will develop and compare four classifiers. We will have two multimodal classifiers, one combining MobileNetV2 image features with scaled patient metadata and one combining ResNet50 image features with scaled patient metadata. Then we will have two image-only classifiers, one using MobileNetV2 image features and the other using ResNet50 image features. All the models will have class weights applied to address the imbalance in classes. And early stopping will also be used, as to not over train the data.

Step 5: Model Evaluation

- We will be evaluating the models' accuracy by using a macro F1 score. Using the macro F1 score calculates the F1 score for each class independently, and then takes the unweighted average. Thus, the classes are treated equally since our dataset is imbalanced. We will then create confusion matrices for each classifier to view the misclassification patterns across the different lesion types.

Step 6: Comparative Analysis

- First we will compare the ResNet50 models with the MobileNetV2 models to see which pretrained model performed better. Then we will compare the performance of the best image-only model and the best multi-modal model to determine whether incorporating patient metadata improves classification accuracy or not. We aim for our models to be able to classify pigmented skin lesions with an f1 score of above 90%. This is best from a medical standpoint, but with the limited amount of images we have and the class imbalance, we understand this might not be possible.



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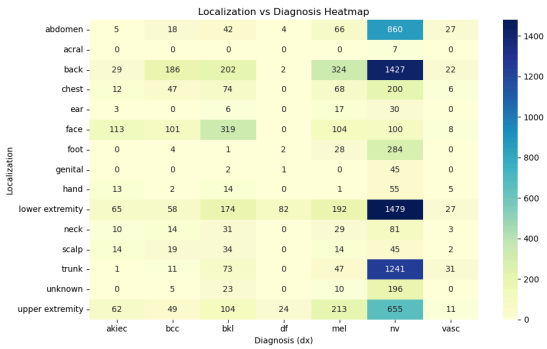

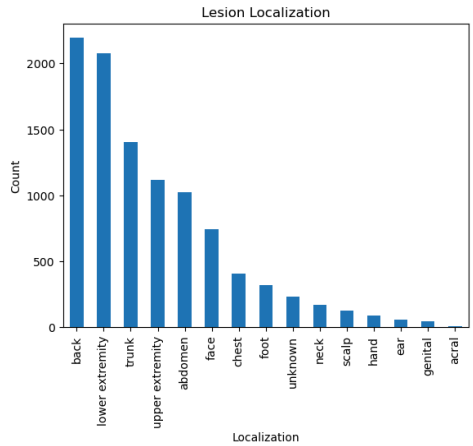

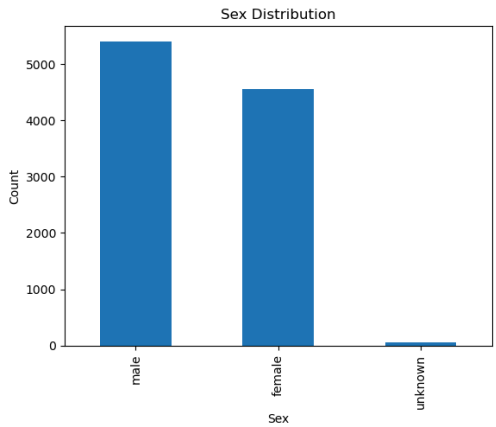

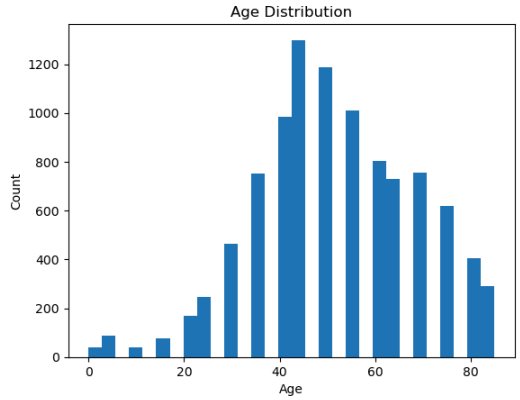

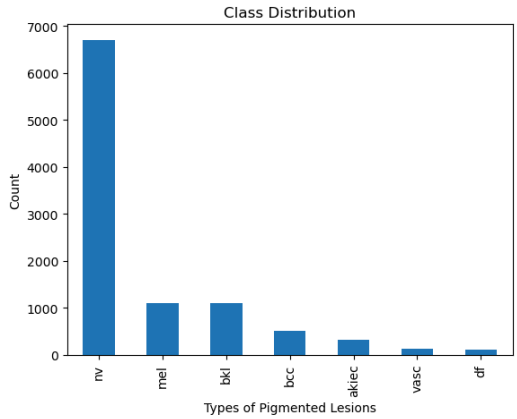

References

[1] P. Tschandl, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." Harvard Dataverse, 2018. doi: 10.7910/DVN/DBW86T.

[2] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," *arXiv.org*, 2018. https://arxiv.org/abs/1801.04381

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[4] I. D. Bassukas and A. Tatsioni, "Male Sex is an Inherent Risk Factor for Basal Cell Carcinoma," *Journal of Skin Cancer*, vol. 2019, pp. 1–5, Oct. 2019, doi: https://doi.org/10.1155/2019/8304271.